

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

TELEVISION CONTROL MODULE (TVCM) - CONDITIONAL ACCESS SYSTEM

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable.
Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Television Control Module (TVCM) - Conditional Access System. For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual. For additional information, refer to: [Information and Entertainment System](#) (415-

00 Information and Entertainment System - General Information, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B147C-00	Conditional Access System (Video Encryption) - No sub type information	NOTE: This fault is detected when the engine is running (Powermode = 7) - Conditional access system card not authorised or expired - Conditional access system card fault	NOTE: This DTC is applicable to Japan market vehicles only. - Subscription required - Install a new conditional access system card
B1A56-11	Antenna - Circuit short to ground	NOTES: - This fault is detected when the engine is running (Powermode = 7) and is checked every 100 milliseconds - Circuit reference - TV1 SIG - - Antenna 1 signal circuit short circuit to ground	Refer to the electrical circuit diagrams and check the antenna 1 signal circuit for short circuit to ground
B1A56-12	Antenna - Circuit short to battery		Refer to the electrical circuit diagrams and check the antenna 1 signal circuit for short circuit to power

		NOTES: ▪ This fault is detected when the engine is started (Powermode = 7) ▪ Circuit reference - TV1 SIG -	
		▪ Antenna 1 signal circuit short circuit to power	
B1A56-13	Antenna - Circuit open	NOTES: ▪ This fault is detected when the engine is running (Powermode = 7) and is checked every 100 milliseconds ▪ Circuit reference - TV1 SIG -	▪ Refer to the electrical circuit diagrams and check the antenna 1 signal circuit for open circuit, high resistance
B1D55-11	Antenna #2 - Circuit short to ground	NOTES: ▪ This fault is detected when the engine is running (Powermode = 7) and is checked every 100 milliseconds ▪ Circuit reference - TV2 SIG -	▪ Refer to the electrical circuit diagrams and check the antenna 2 signal circuit for short circuit to ground

		Antenna 2 signal circuit short circuit to ground	
B1D55-12	Antenna #2 - Circuit short to battery	NOTES: - This fault is detected when the engine is started (Powermode = 7) - Circuit reference - TV2 SIG - - Antenna 2 signal circuit short circuit to power	Refer to the electrical circuit diagrams and check the antenna 2 signal circuit for short circuit to power
B1D55-13	Antenna #2 - Circuit open	NOTES: - This fault is detected when the engine is running (Powermode = 7) and is checked every 100 milliseconds - Circuit reference - TV2 SIG - - Antenna 2 signal circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the antenna 2 signal circuit for open circuit, high resistance
B1D56-11	Antenna #3 Circuit - Circuit short to ground		Refer to the electrical circuit diagrams and check the antenna 3 signal circuit for short circuit to ground

		NOTES: ▪ This fault is detected when the engine is running (Powermode = 7) and is checked every 100 milliseconds ▪ Circuit reference - TV3 SIG -	
		▪ Antenna 3 signal circuit short circuit to ground	
B1D56-12	Antenna #3 Circuit - Circuit short to battery	NOTES: ▪ This fault is detected when the engine is started (Powermode = 7) ▪ Circuit reference - TV3 SIG -	▪ Refer to the electrical circuit diagrams and check the antenna 3 signal circuit for short circuit to power
B1D56-13	Antenna #3 Circuit - Circuit open	NOTES: ▪ This fault is detected when the engine is running (Powermode = 7) and is checked every 100 milliseconds ▪ Circuit reference - TV3 SIG -	▪ Refer to the electrical circuit diagrams and check the antenna 3 signal circuit for open circuit, high resistance

		Antenna 3 signal circuit open circuit, high resistance	
B1D57-11	Antenna #4 Circuit - Circuit short to ground	NOTES: - This fault is detected when the engine is running (Powermode = 7) and is checked every 100 milliseconds - Circuit reference - TV4 SIG - - Antenna 4 signal circuit short circuit to ground	Refer to the electrical circuit diagrams and check the antenna 4 signal circuit for short circuit to ground
B1D57-12	Antenna #4 Circuit - Circuit short to battery	NOTES: - This fault is detected when the engine is started (Powermode = 7) - Circuit reference - TV4 SIG - - Antenna 4 signal circuit short circuit to power	Refer to the electrical circuit diagrams and check the antenna 4 signal circuit for short circuit to power
B1D57-13	Antenna #4 Circuit - Circuit open		Refer to the electrical circuit diagrams and check the antenna 4 signal circuit for open circuit, high resistance

		NOTES: ▪ This fault is detected when the engine is running (Powermode = 7) and is checked every 100 milliseconds ▪ Circuit reference - TV4 SIG -	
		▪ Antenna 4 signal circuit open circuit, high resistance	
B1D58-11	Television Output - Circuit short to ground	NOTES: ▪ This fault is detected when the engine is running (Powermode = 7) and is checked every 200 milliseconds ▪ Circuit reference - CVBS OUT - ▪ CVBS signal circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the CVBS signal circuit for short circuit to ground
B1D58-12	Television Output - Circuit short to battery		▪ Refer to the electrical circuit diagrams and check the CVBS circuit for short circuit to power

		NOTES: ▪ This fault is detected when the engine is running (Powermode = 7) and is checked every 200 milliseconds ▪ Circuit reference - CVBS OUT -	
		▪ CVBS signal circuit short circuit to power	
B1D58-13	Television Output - Circuit open	NOTES: ▪ This fault is detected when the engine is running (Powermode = 7) and is checked every 200 milliseconds ▪ Circuit reference - CVBS OUT -	▪ Refer to the electrical circuit diagrams and check the CVBS circuit for open circuit, high resistance
U2100-00	Initial Configuration Not Complete - No sub type information	NOTE: This fault is detected when the engine is started (Powermode = 7) ▪ Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs.

		Incorrect television control module installed	- Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary. Clear the DTCs and retest - Install a new television control module as necessary
U2101-00	Control Module Configuration Incompatible - No sub type information	NOTE: This fault is detected when the engine is started (Powermode = 7) - Car configuration file mismatch with vehicle specification - Incorrect television control module installed	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs. - Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary. Clear the DTCs and retest - Install a new television control module as necessary
U3000-46	Control Module - Calibration/parameter memory failure	NOTE: This fault is detected when the engine is started (Powermode = 7) Television control module is not configured correctly	Using the manufacturer approved diagnostic system, re-configure the television control module with the latest level software

U3000-49	Control Module - Internal electronic failure	NOTE: This fault is detected when the engine is running (Powermode = 7) and is checked every 200 milliseconds Television control module internal failure	Using the manufacturer approved diagnostic system, re-configure the television control module with the latest level software. Clear the DTCs and retest. If the fault persists, install a new television control module
U3000-98	Control Module - Component or system over temperature	NOTE: This fault is detected when the engine is running (Powermode = 7) and is checked every 1000 milliseconds Television control module over temperature	Check the airflow to the television control module. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new television control module
U3003-62	Battery Voltage - Signal compare failure	NOTE: This fault is detected when the engine is running (Powermode = 7) and is checked when the GlobalSnapShotData is updated Mismatch between the voltage at the television control module and the voltage value broadcast on the CAN bus and MOST bus	Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111) - and compare it to battery voltage. Refer to the electrical circuit diagrams and check the television control module power and ground circuits for open circuit, high resistance

